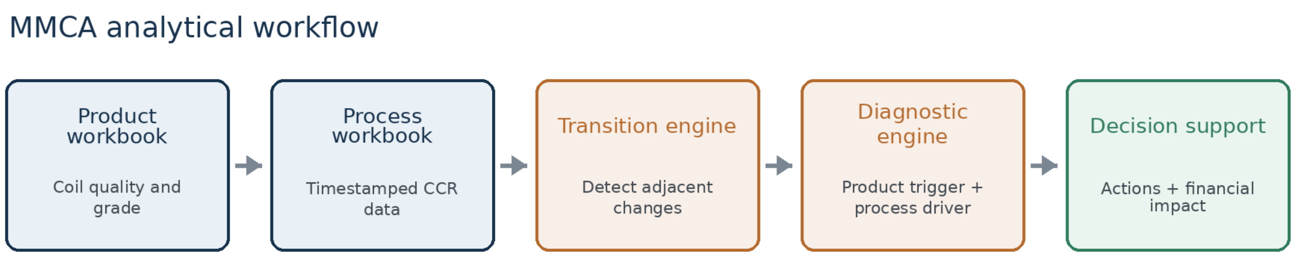
MMCA

Grade Transition Intelligence

User and Methodology Guide

Continuous-cast copper rod grade-change analysis, root-cause attribution and financial decision support



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This guide contains methodology and user instructions only. It does not contain company production data.

Contents

Quick reading guide

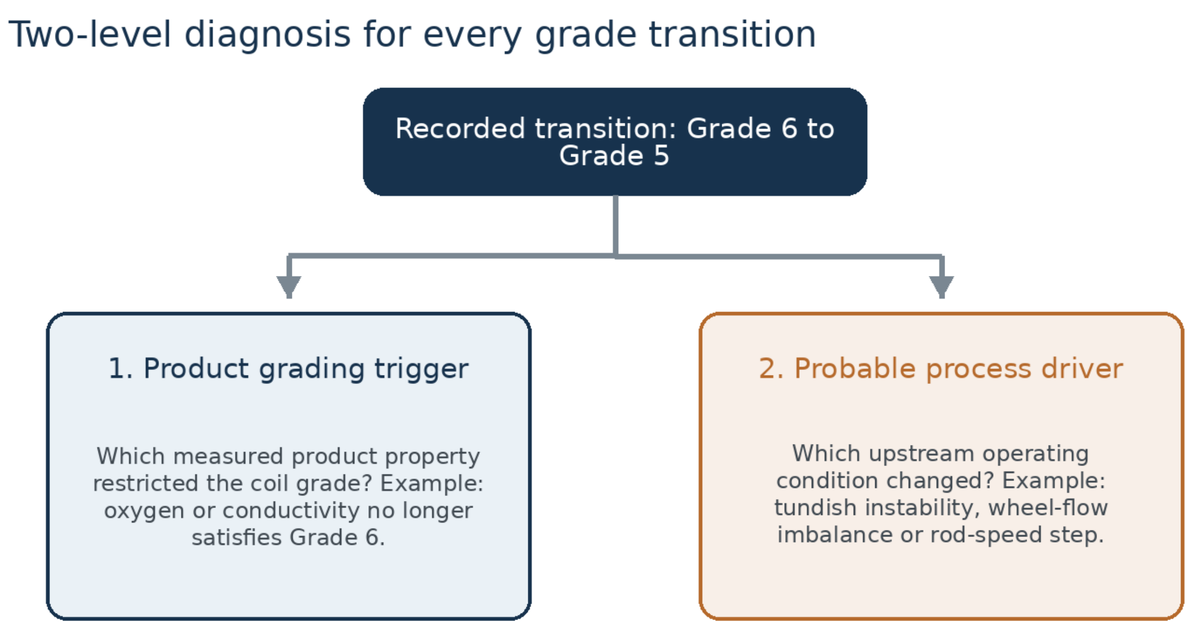
Operational users can focus on Sections 1, 2, 5, 9, 10, 12 and 13. Data analysts and process engineers should also read Sections 6, 7 and 8.

1. Executive summary

MMCA Grade Transition Intelligence is a reproducible Streamlit application for analysing sudden changes in the grade of continuous-cast copper rod. It accepts the monthly product-quality workbook and the timestamped process workbook supplied by the company, reconstructs the production sequence, detects every adjacent grade change and generates an explainable Quality Control report.

The application separates two questions that are often mixed together:

1. Product grading trigger: Which measured product property explains why the new coil received its recorded grade?
2. Probable process driver: Which upstream operating condition changed before or during the affected production window?



Decision-support boundary
The application identifies statistical association and engineering plausibility. It does not automatically prove causation and does not replace the official grading procedure, operator judgement or engineer review.

The application provides four levels of analysis:

Analytics layer	Question answered	Main outputs
Descriptive	What happened?	Transition timeline, downgrade and upgrade counts, persistence, affected tonnage
Diagnostic	Why did it happen?	Product triggers, process deviations and ranked probable drivers

Analytics layer	Question answered	Main outputs
Prescriptive	What should be done?	Prioritised QC and process actions with verification metrics
Financial	What is the commercial exposure?	Estimated loss, potential savings and net revenue protected

2. Problem statement and analytical framing

Copper rod is manufactured continuously. A coil is created only when the rod is cut according to a customer requirement. A sudden change between two consecutive coils can therefore interfere with customer allocation, create off-plan inventory and increase rework, remelting or metal wastage.

$$\text{Grade-transition event: } G_i \neq G_{i-1}$$

G_i is the current coil grade and G_{i-1} is the immediately preceding coil grade.

Why the transition is the analytical unit

An isolated Grade 4 coil does not reveal whether it was planned. A Grade 5 to Grade 4 change, followed by several Grade 4 coils, defines a production event with a start time, persistence, affected weight and business impact.

The app classifies transitions as:

- Upgrade: current grade is higher than the previous grade.
- Downgrade: current grade is lower than the previous grade.
- One-coil fluctuation: the changed grade lasts for only one coil.
- Sustained transition: the new grade persists for several coils.
- Special-context event: startup, stoppage, manual grade, contamination or another operational remark.

3. Application inputs and analysis scope

3.1 Product and coil-quality workbook

The product workbook contains the coil sequence, recorded grade, production time, weight, quality measurements and remarks. The application automatically detects the Metrod Master or Master_Simplified worksheet and searches the first rows for the true column header.

Data group	Examples
Identification and time	Coil ID, internal coil ID, date, start time, finish time
Commercial/production	Recorded grade, coil weight, order or client fields when available
Mechanical and electrical	Tensile strength, elongation, conductivity, twist tests
Surface and oxygen	Defect counts, oxygen, total oxide
Chemistry	Al, Sb, As, Bi, Cd, Fe, Pb, Ni, P, Se, S, Te, Sn and Zn

Data group	Examples
Context	Startup, stoppage, manual-grade, contamination and other remarks

3.2 Process workbook

The process workbook contains timestamped CCR measurements. The application focuses on physically interpretable variables that can reasonably influence rod grade.

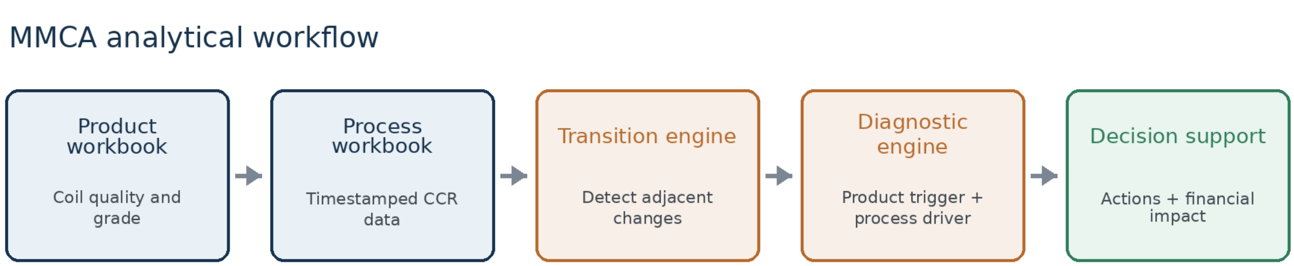
Process group	Parameters considered
Thermal control	SV, HF, tundish, wheel, bar-entry and rolling-mill-entry temperatures
Casting and line motion	Rod speed, HF weight and documented casting indicators
Cooling-flow balance	Wheel, band, side, bottom, nip-nozzle, after-cooler and stripper-shoe flows
Lubrication and treatment	Soluble-oil temperature/concentration, IPA, NAPS flow/concentration and wax concentration

Excluded channels
Undocumented SCU colour-coded channels are not ranked until the company confirms their physical meaning. This avoids producing misleading causal interpretations.

3.3 Analysis scope

The user can analyse all uploaded data or select a start date and end date. Only coils in the chosen production range are included. Process data are filtered to cover the same period, with additional time around the boundaries so that process-to-coil alignment remains valid.

4. Data analytics workflow



1. Load and validate the product and process workbooks.
2. Combine production date and time fields and sort coils chronologically.
3. Detect every adjacent grade transition and calculate the persistence of the new grade.

4. Evaluate product measurements against the applicable Revision 19 or Revision 20 grading criteria.
5. Map timestamped process records to each coil-production window.
6. Compare the affected coil with the preceding stable operating condition.
7. Rank probable process drivers using robust deviation, step change, completeness and engineering relevance.
8. Aggregate contribution percentages and generate actionable recommendations.
9. Estimate financial exposure and prospective savings using editable assumptions.
10. Generate a printable HTML report and an Excel evidence pack.

5. Descriptive analytics and transition detection

5.1 Chronological reconstruction

The application rebuilds the production order using the coil start time, finish time and coil identifier. This is essential because spreadsheet row order cannot be assumed to represent the actual production sequence.

5.2 Transition metrics

- Total coils analysed and date range.
- Total transitions and transitions per 100 coils.
- Number and percentage of downgrades and upgrades.
- Number and percentage of one-coil fluctuations.
- Most frequent grade pairs, such as Grade 5 to Grade 4.
- Persistence of each changed grade run.
- Affected production weight for each transition.
- Operational context of startup, stoppage, contamination and manual grading.

5.3 Affected production weight

The affected weight is the combined weight of the continuous run that starts with the changed grade. It is not limited to the first changed coil.

$$W_{affected} = \Sigma\, W_j \quad \text{for all coils in the changed-grade run}$$

Example: if Grade 5 changes to Grade 4 and the following three coils remain Grade 4, the event weight is the sum of all three Grade 4 coils. This better represents the inventory and customer-allocation impact.

6. Product grading and product-trigger analysis

6.1 Parameter-level grading

Each available product measurement is independently assigned the highest grade whose approved criterion it satisfies. The grading revision is selected according to production date.

$$G_{computed} = \min(G_1, G_2, G_3, \ldots, G_n)$$

The most restrictive valid parameter grade determines the computed product grade.

Parameter	Measured value	Parameter grade
Oxygen	320 ppm	5
Total oxide	180 Å	6
Conductivity	101.1% IACS	5
25 RTF twist	32	6

In the example, the computed grade is Grade 5 and the limiting parameters are oxygen and conductivity.

6.2 Product transition triggers

For each grade transition, the application compares the previous and current parameter-level grades. A parameter is considered a strong trigger when its grade changes in the same direction as the recorded coil grade, its current parameter grade is close to the recorded grade and the measurement is valid.

Interpretation
A product trigger explains why the coil was classified at a particular grade. It is a product-level explanation, not necessarily the upstream process cause.

7. Process-to-coil alignment and data quality

7.1 Influence window

The product table is coil-based, while the process table contains frequent timestamped measurements. The app assigns process measurements to a coil using its production window. A configurable process lag can shift the influence window earlier when actual material residence time is known.

$$t_{start,influence} = t_{coil,start} - t_{lag}$$

$$t_{finish,influence} = t_{coil,finish} - t_{lag}$$

7.2 Coil-level process features

- Number of valid measurements
- Mean, minimum and maximum
- Standard deviation and range
- First and last value
- Within-coil change

7.3 Data-quality screening

Industrial data can contain missing readings, constant or frozen channels, duplicate timestamps, sensor spikes and physically impossible values. The application screens each numerical channel and excludes or penalises unreliable evidence.

Check	Reason
Missing-value percentage	Low coverage reduces confidence in an event-level diagnosis
Unique-value count	Detects constant or frozen channels
Minimum, maximum and percentiles	Highlights impossible or extreme readings
Timestamp consistency	Prevents incorrect process-to-coil mapping
Sample count in the coil window	Ensures the event contains enough evidence

8. Process root-cause scoring

8.1 Local stable baseline

For every transition, the current coil is compared with the immediately preceding coil and a local baseline of preceding stable coils. The default baseline is five coils, with preference given to coils of the previous grade.

8.2 Robust deviation

$$Z_{robust} = |x_{current} - \text{median}(x_{baseline})| / [1.4826 \times \text{MAD}(x_{baseline})]$$

The median absolute deviation is less sensitive to isolated spikes than ordinary standard deviation, making it suitable for noisy industrial data.

8.3 Immediate step change

$$Z_{step} = |x_{current} - x_{previous}| / \text{historical robust scale}$$

This component captures sudden changes between consecutive coils, even when the value remains inside a broad operating range.

8.4 Engineering relevance

The score also accounts for whether a process change is physically relevant to the product trigger. For example, oxygen and oxide deterioration increase the relevance of thermal, casting-speed and cooling variables, while surface defects increase the relevance of cooling balance and lubrication/NAPS conditions.

8.5 Candidate score

$$S = 100 \times [0.45 \cdot \min(Z_{robust}/4,1) + 0.25 \cdot \min(Z_{step}/4,1) + 0.15 \cdot C + 0.15 \cdot R]$$

Score component	Weight	Interpretation
Deviation from stable baseline	45%	How unusual the current coil is relative to recent stable production
Immediate coil-to-coil change	25%	How sharply the parameter changed at the transition
Data completeness	15%	Whether enough valid measurements support the comparison
Product-process relevance	15%	Whether the parameter is physically connected to the product trigger

Score range	Confidence band
75 to 100	High
50 to below 75	Moderate
Below 50	Low

Important
Candidate scores are explainable attribution scores. They indicate association and engineering plausibility, not confirmed causation.

9. Contribution percentages and recommendations

9.1 Event-level contribution

For each transition, retained candidate scores are normalised so that their event-level contributions sum to 100%.

$$c_{e,j} = S_{e,j} / \Sigma S_{e,k}$$

9.2 Tonnage weighting

Each event contribution is multiplied by the affected run weight. This prevents many small one-coil events from automatically dominating fewer but larger production losses.

$$w_{e,j} = c_{e,j} \times W_e$$

9.3 Overall contribution

$$C_j = [\Sigma_e w_{e,j} / \Sigma_e \Sigma_k w_{e,k}] \times 100$$

The dashboard reports contribution by process group and individual parameter. These percentages are model attribution weights, not experimentally measured causal fractions.

9.4 Actionable recommendations

Leading group	Typical actions	Verification metric
Thermal control	Check thermocouple calibration, setpoint-versus-actual drift, launder heat loss and alarm limits	Lower temperature deviation and fewer thermal-associated transitions
Cooling-flow balance	Inspect nozzles, filters, pumps and valves; compare machine-side and operator-side flow	Reduced side-to-side imbalance and fewer surface-related downgrades
Casting and line speed	Review abrupt rod-speed changes, encoder condition and controlled ramp limits	Fewer transitions following speed steps
Lubrication and NAPS	Tighten oil, IPA, NAPS and wax checks; compare online and laboratory values	Improved concentration stability and fewer surface-quality events
Data quality	Repair frozen channels, impossible readings and timestamp issues	Higher valid-data coverage and fewer low-confidence diagnoses

10. Financial impact model

Financial interpretation
The financial module is an editable opportunity-loss scenario. It is not an audited accounting calculation because planned grade, realised selling price, rework cost and recovery values are not fully structured in the uploaded files.

10.1 Temporary target grade

For a downgrade, the preceding stable grade is temporarily treated as the intended commercial grade. The current model monetises downgrades only. Unexpected upgrades can also create off-plan inventory, but they are not assigned a loss in Version 1.

10.2 Copper benchmark

The application can retrieve a historical daily COMEX HG copper benchmark and a USD/MYR exchange rate. Manual values are used as fallback or can be entered deliberately.

$$P_{USD/MT} = P_{USD/lb} \times 2204.6226$$

$$P_{MYR/MT} = P_{USD/MT} \times R_{USD/MYR}$$

10.3 Grade steps lost

$$L_g = G_{previous} - G_{current} \quad \text{for a downgrade}$$

10.4 Gross copper value exposed

$$V_{gross} = W_{affected} \times P_{MYR/MT}$$

10.5 Commercial grade loss

$$L_{commercial} = V_{gross} \times L_g \times (d / 100)$$

d is the commercial penalty percentage per grade step.

10.6 Additional Grade 0 loss

$$L_{remelt} = V_{gross} \times (r / 100) \quad \text{when the new grade is Grade 0}$$

10.7 Total estimated loss

$$L_{estimated} = L_{commercial} + L_{remelt}$$

$$L_{total} = \Sigma\, L_{estimated,e}$$

10.8 Average loss per downgrade

$$L_{average} = L_{total} / N_{downgrades}$$

10.9 Potential savings

$$S_{potential} = L_{total} \times (p / 100) \times (e / 100)$$

p is preventable share and e is expected intervention effectiveness.

10.10 Potential net revenue protected

$$R_{protected} = S_{potential} - C_{implementation}$$

Editable assumption	Default placeholder
Fallback copper price	USD 10,000 per metric tonne
Fallback USD/MYR exchange rate	4.50
Commercial penalty per grade step	0.50%
Additional Grade 0 recovery loss	3.00%
Preventable share	50%
Expected intervention effectiveness	60%

Editable assumption	Default placeholder
Implementation cost	MYR 0

11. Worked financial example

Input	Value
Transition	Grade 5 to Grade 3
Affected run weight	20 MT
Copper price	USD 10,000/MT
USD/MYR	4.50
Penalty per grade step	0.50%
Preventable share	50%
Intervention effectiveness	60%
Implementation cost	MYR 1,000

$$\text{Copper price in MYR/MT} = 10{,}000 \times 4.50 = \text{MYR } 45{,}000$$

$$\text{Gross copper value} = 20 \times 45{,}000 = \text{MYR } 900{,}000$$

$$\text{Grade steps lost} = 5 - 3 = 2$$

$$\text{Estimated loss} = 900{,}000 \times (2 \times 0.50\%) = \text{MYR } 9{,}000$$

$$\text{Potential savings} = 9{,}000 \times 50\% \times 60\% = \text{MYR } 2{,}700$$

$$\text{Net revenue protected} = 2{,}700 - 1{,}000 = \text{MYR } 1{,}700$$

Financial metric	Result
Gross copper value exposed	MYR 900,000
Estimated grade loss	MYR 9,000
Potential savings	MYR 2,700
Potential net revenue protected	MYR 1,700

12. How to use the application

1. Open the Streamlit application.

2. Upload the product workbook: CCR Coil Inspection details Master.

3. Upload the process workbook: Process_Values.

4. Confirm that both files are validated.

5. Choose All uploaded data or Select date range.

6. Leave the process-to-coil lag at zero unless the actual residence time is known.

7. Choose the number of preceding stable coils used as the baseline. The default is five.

8. Choose the minimum root-cause score. The default is 25.

9. Click Run grade-change analysis.

10. Review the Overview, Transition investigator, Root causes and actions, and Financial impact tabs.

11. Enter company-approved financial assumptions before using monetary outputs in decision-making.

12. Download the HTML report and Excel evidence pack.

12.1 Recommended first use

- Start with a short date range to confirm that coil order and process overlap are correct.
- Inspect several known transitions with QC and production engineers.
- Confirm that the top-ranked parameters are physically meaningful.
- Calibrate the process lag before comparing results across months.

13. How to interpret the outputs

Output	Correct interpretation	Incorrect interpretation
Product trigger	The measurement that restricted the recorded grade	The proven upstream cause
Candidate score	Strength of statistical and engineering evidence	Probability that the parameter caused the event
Contribution percentage	Relative attribution weight in the model	Experimentally measured causal share
Estimated loss	Scenario-based commercial exposure	Audited financial loss
Potential savings	Prospective recoverable value under assumptions	Savings already realised
Net revenue protected	Potential value after implementation cost	Current-period booked revenue

Best practice
Use the event investigator to connect each conclusion to a transition ID, coil pair, product trigger, process trace and data-quality status before taking action.

14. Limitations, validation and responsible use

14.1 Current limitations

- Raw-material causation requires timestamped cathode, scrap and charge-mix records.

- The process lag must be calibrated using actual line distance and rod speed.
- Manual-grade, startup, stoppage and contamination events require human review.
- Financial estimates require company-approved grade realisation, rework and recovery assumptions.
- Unexpected upgrades are not monetised in the current financial model.
- The app does not replace the approved grading procedure or operator judgement.

14.2 Recommended validation process

1. Select historical transitions for which the cause is already known.
2. Ask QC and production engineers to record the confirmed primary and secondary cause.
3. Compare confirmed causes with the application's top one and top three candidates.
4. Tune baseline length, process lag, relevance weights and score threshold.
5. Track top-one and top-three precision over time.
6. Implement selected actions and monitor transition rate per 100 coils, affected tonnage and estimated loss.

14.3 Data confidentiality

The source repository should not contain company workbooks or generated reports containing coil, customer or process data. For cloud deployment, use a private repository and restricted application access. Use local deployment when company policy does not permit cloud processing.

15. Glossary of key terms

Term	Meaning
Affected weight	Combined weight of the continuous run that starts with a grade transition
Baseline coils	Recent stable coils used as the local reference condition
Candidate score	0-100 attribution score combining deviation, step change, completeness and relevance
Computed product grade	Most restrictive grade supported by the available product measurements
Contribution percentage	Normalised, tonnage-weighted attribution weight
Downgrade	A transition where the current grade is lower than the previous grade
Event context	Operational label such as normal production, startup, stoppage or manual grade
Product trigger	Product measurement that explains the new grade classification
Process driver	Upstream process condition that may have contributed to the product change
Process lag	Time correction between an upstream measurement and the affected coil material
Robust z-score	Deviation measure based on the median and median absolute deviation
Transition persistence	Number of consecutive coils produced at the new grade

Term	Meaning
Upgrade	A transition where the current grade is higher than the previous grade

Conclusion

MMCA Grade Transition Intelligence converts two routine company workbooks into a traceable decision-support workflow. It maps every sudden grade change, explains the product measurements behind the grade, ranks plausible upstream process contributors, proposes actions and estimates the financial value of reducing avoidable fluctuations.

Its purpose is to support structured collaboration between Quality Control, Production, Process Engineering and Finance while keeping assumptions transparent and conclusions reviewable.